

CHECKLIST

SMALL CRAFT - Electric propulsion systems

Ref.: EN ISO 16315:2016

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	



Installation documentation as per Annex B

Subject to check	Clause	Requirements	Checked ?
1 For the battery group data including the following: - battery technology/type; - prospective short circuit current in (kA) and time constant in (s); - ventilation and temperature parameters; - Ampere hours (Ah).	Annex B	[Yes ?]	
2 Generator data including the following: - power (kW), rated voltage, power factor, frequency, other; - necessary reactance's and time constants; - maximum operating temperature.	Annex B	[Yes / NA ?]	
3 Propulsion motor data including the following: - power (kW), rated voltage, power factor, frequency, other; - regeneration performances; - braking data; - maximum operating temperature.	Annex B	[Yes ?]	
4 Charger data including the following: - power (kW), rated voltage, power factor, frequency, other; - short-circuit current (kA); - maximum operating temperature.	Annex B	[Yes ?]	

Further documentation

5 Overcurrent protection for each propulsion unit: position and rating		[Yes ?]	
6 Wire diagram including conductor length and type		[Yes ?]	
7 Voltage drop calculation	10.6	[Yes ?]	
8 Proof of separation of the DC system for propulsion and household [see RCD, Annex I, para 5.3]		[Yes ?]	
9 Proof that semiconductor converters shall conform to IEC 60146, if applicable	4.7.2	[Yes ?]	
10 Proof that the electrical motor complies with IEC 60034	4.7.3	[Yes ?]	
11 Information about the identification of all electrical equipment and enclosures	4.9.1	[Yes ?]	
12 Proof of compliance for an electric/electronic throttle according to EN ISO 25197:2018		[Yes / NA ?]	
13 EMC compliance with IEC 60945 for the electronic equipment	4.12	[Yes ?]	
14 Ignition protection of electrical equipment in the same vicinity as a battery / battery pack if emitting flammable gas or substances is possible		[Yes ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

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15	Restrictions of hazardous areas and its accessibility, if applicable	4.14	[Yes ?]	
16	Information and proof about fault tripping of controls, its reset and emergency stop and manual emergency control mode	5.1.1	[Yes ?]	
17	Information and proof for instruments, alerts and trip alarm	5.2	[Yes ?]	
18	Information and proof for protection against electrical shock: accidental contact; insulation resistance of 500 ohm/V	6.1.3	[Yes ?]	
19	Information and proof for fault to earth automatic disconnection	6.2, 6.3	[Yes ?]	
20	Information and proof of the overcurrent protection, type and characteristic; see IEC 60947		[Yes ?]	
21	Information about the battery monitoring and installation	8	[Yes ?]	
22	Documentation about the compliance of the electrical installation which refers to IEC 60092-352 and other standards	9	[Yes ?]	
23	Earthing and bonding testing	10.2	[Yes ?]	
24	Insulation resistance of all components in the system.	10.3	[Yes ?]	
25	On load test and inspection of electrical propulsion systems, and associated switch gear and control gear	10.5	[Yes ?]	
26	On load test; Voltage drop;	10.6	[Yes ?]	

Instructions/Warnings to be included in the owner's manual

Subject to check	Clause	Requirements	Checked ?
27 Block diagram of the propulsion system and a description of the installation location of the electric assemblies;conditions.	A.1	[Yes ?]	
28 Diagram(s) identifying the electrical circuits of the craft with the locations of electrical devices in the craft and identification of conductors by colour or other means;	A.1	[Yes ?]	
29 Location and description of functions of electrical controls, dials, switches, fuses and also circuitbreakers installed on the panel board;	A.1	[Yes ?]	
30 Instructions on how to leave the crafts electrical propulsion system when unattended;	A.1	[Yes ?]	
31 Description of display information as required by Clause 5;	A.1	[Yes ?]	
32 Marine installation, alterations and maintenance must be performed by a competent marine electrical technician.	A.1	[Yes ?]	
33 Manufacturer instructions for any electrical machinery provided for consumer use	A.1	[Yes ?]	
34 Following advisory instructions are provided for the owner's manual to never: - work on the electrical installation while the system is energised; - modify the propulsion system, battery type and system components; - alter or modify the rated current amperage of overcurrent protective devices; - install or replace electrical appliances or devices with components which exceed the rated current amperage of the circuit.	A.2	[Yes ?]	